

altairsim — Color graphics on a Cromemco Dazzler

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Four machines, each with a **Cromemco Dazzler** in an Altair:

Machine file	What it is
<code>kscope.toml</code>	Li-Chen Wang's Kaleidoscope , an endless four-way-mirrored pattern. It comes up drawing.
<code>dzmbasic.toml</code>	DZMBASIC , Microsoft BASIC with Dazzler graphics, under CP/M 2.2. You draw with <code>DZOP</code> and <code>DZF</code> . See DZMBASIC under CP/M .
<code>games.toml</code>	The Dazzler games under CP/M 2.2, with a Cromemco D+7A for the joysticks and the sound. See The games under CP/M .
<code>gdemo.toml</code>	Cromemco's GDEMO , a demonstration of Dazzler graphics. It comes up drawing. See GDEMO .

Kaleidoscope

Li-Chen Wang's **Kaleidoscope** (`KSCOPE`) — the classic Dazzler demo — drawing a four-way-mirrored pattern that wanders and recolors forever.

```
cd examples/dazzler
altairsim kscope.toml
```

The machine comes up **drawing**: `kscope.toml` loads `KSCOPE` and RUNs it from its `startup` list, on an Altair that has a **Dazzler** in it — a Z80 at 4 MHz, 64K of RAM, an 88-2SIO for the console, and the Dazzler's two ports at `0E` / `0F`. On a build with **SDL3** a window opens the moment the Dazzler turns on, and the kaleidoscope appears: a 2 KB, 64×64, 16-color picture.

The processor is a `z80` board, at the 4 MHz a Cromemco Z-1 ran. `KSCOPE` itself is 8080 code and does not care — a Z80 runs it unchanged — but it is the CPU the rest of the period Dazzler software wants, and fitting one is a single line in the machine file.

`KSCOPE` never stops on its own (there is no `HLT`), so press **STOP** (`Ctrl-E`) at the terminal to break back to the `altairsim>` prompt; `RUN 0` starts it again. On a **headless** build the machine runs exactly the same and simply draws nothing.

To start it by hand instead — for instance to watch it draw into memory — break out with STOP and re-run it yourself:

```
altairsim> LOAD KSCOPE.HEX
loaded 127 bytes (1 page) from KSCOPE.HEX (0000-007E)
altairsim> RUN 0
```

What KSCOPE does

The program (`KSCOPE.ASM` , with its assembler listing in `KSCOPE.PRN`) is tiny and assembles at `0000` , so it runs straight from a `RUN 0` :

- It turns the Dazzler **on** with a framebuffer at `0200` (`OUT 0EH = 81h`), and sets the format to **2 KB, 64×64, color** (`OUT 0FH = 30h`).
- The 2 KB picture is four 512-byte **quadrants** tiled 2×2. KSCOPE draws one pixel and then mirrors it into all four quadrants by negating each axis — which is why the pattern is symmetric about both the horizontal and vertical center. It walks a pseudo-random path and cycles the color, so the figure is always moving.

Because the framebuffer starts as whatever the RAM powered up holding (random, like real static RAM), the picture emerges from a field of color noise as KSCOPE paints over it.

The Dazzler, briefly

The Dazzler reads its picture straight out of **main memory** — the framebuffer is ordinary RAM (here at `0200`), not on the card. Two `OUT` ports drive it:

- `OUT 0EH` — control: bit 7 on/off, bits 6–0 the high address bits of the framebuffer (so the base is 512-byte aligned).
- `OUT 0FH` — format: resolution (32×32...128×128), size (512 B or 2 KB), color vs 16 greys.

`IN 0EH` reads two status bits (odd/even scan line, end-of-frame) a program can poll to pace its drawing to the frame.

Try it yourself

- **Watch it draw into memory.** Break out (`Ctrl-E`) and `DUMP 0200` — the framebuffer bytes KSCOPE has written are right there; each byte is two 4-bit color elements.
- **Change the colors.** The picture is a palette machine: the same bytes look different under a different `OUT 0FH` . `SET daz0` shows the card; the format is set by the running program.

- **Slow it down or speed it up.** `SET cpu0 clock_hz=2000000` for a 2 MHz Altair, or `clock_hz=0` for flat out.

DZMBASIC under CP/M

`dzmbasic.toml` boots CP/M 2.2 off the 8" floppy in `DZMBASIC.DSK`. That disk holds `DZMBASIC.COM`, a Microsoft BASIC with two additions for the Dazzler: the `DZOP` statement and the `DZF` function. Launch it and start the program:

```
cd examples/dazzler
altairsim dzmbasic.toml
```

```
56K CP/M
Version 2.2mits (07/28/80)

A>DZMBASIC
BASIC-80 Rev. 5.21 (DZHT)
[CP/M Version]
Copyright 1977-1981 (C) by Microsoft
20251 Bytes free
Ok
```

The machine boots itself and comes up at `A>`. On a build with **SDL3**, a window opens the first time a program touches the Dazzler. Try it:

```
DZOP "I"
DZOP "P",5,5
DZOP "L",50,50
DZOP "S","HI"
PRINT DZF(5,5)
```

`DZOP "I"` initializes the Dazzler, `"P"` moves the cursor, `"L"` draws a line to the point you give, and `"S"` writes text. `DZF(x,y)` reads a dot back and gives its color code. The last line prints 1 here, because the line starts at (5,5). `SYSTEM` leaves BASIC and returns to `A>`. **DZMBASIC-Manual.pdf** in this directory lists every `DZOP` action: dots, lines, filled rectangles, text, clear, wait, page flip and palette cycle.

When a statement draws, BASIC can miss the next key you type if you type very fast. Type at a normal pace and the keys are not lost.

The processor is a Z80. The graphics library in DZMBASIC uses Z80 instructions. On an 8080 the first `DZOP "I"` hits an opcode the 8080 does not have, and the machine warm-boots back to `A>`.

`dzmbasic.toml` is the built-in `default` Altair with its 8080 replaced by a Z80, the 88-DCDD holding `DZMBASIC.DSK` in drive 0, and a Dazzler at ports `0E / 0F`.

There is no undo. Drive 0 is read/write, so `SAVE` writes onto the image. Copy it first if you are about to save programs you care about.

The games under CP/M

`games.toml` boots CP/M 2.2 off the 8" floppy in `GAMES.DSK`. That disk holds the Dazzler games.

```
cd examples/dazzler
altairsim games.toml
```

```
56K CP/M
Version 2.2mits (07/28/80)
Copyright 1980 by Burcon Inc.

A>DIR
A: 4DTICTAC COM : CHASE      COM : DAZOFF   COM : DAZON   COM
A: CHESS      COM : DMATION  COM : DOGFIGHT COM : DOODLE   COM
A: GOTCHA     COM : KSCOPE   COM : LIFE    COM : SKETCH  COM
A: SPACEWAR   COM : TANKWAR  COM : TRACK   COM : WRITER  COM
A: XLIFE      COM : R        COM : W        COM : HDIR    COM
A: LS         COM
A>
```

To play a game:

1. Type the name of a game at `A>`, for example `CHASE`, and press Return.
2. Press **STOP** (`Ctrl-E`) to go back to the `altairsim>` prompt when the game has no exit.
3. Type `RUN FF00` to boot CP/M again.

On a build with **SDL3**, a window opens the first time a game turns the Dazzler on. **Cromemco Dazzler Games - Instruction Manual (1978).pdf** in this directory describes the Cromemco games.

The joysticks are on a Cromemco D+7A. The games read two JS-1 joystick consoles through the D+7A board at ports `18` to `1F`. Each console has a joystick, four buttons and a speaker. Each console takes a gamepad if the host has one, and the keyboard if not. At the `altairsim>` prompt, `SHOW d7a0` shows what each console and each speaker is connected to.

The processor is a Z80 at 2 MHz. The crystal gives the games a fixed speed. The speakers of the D+7A play only when the machine has a crystal.

`games.toml` is the built-in `default` Altair with three changes. The 8080 is replaced by the Z80. The 88-DCDD holds `GAMES.DSK` in drive 0. A Dazzler at ports `0E / 0F` and the D+7A are added.

There is no undo. Drive 0 is read/write. Copy `GAMES.DSK` first if you are about to write files onto it.

GDEMO

`gdemo.toml` loads Cromemco's **GDEMO** and RUNs it from its `startup` list. GDEMO is the "Dazzler Graphics Demo Program for Z80 Interface" (1978, version 00.08).

```
cd examples/dazzler
altairsim gdemo.toml
```

```
startup> LOAD GDEMO.HEX
loaded 2691 bytes (12 pages) from GDEMO.HEX (0005-0B7F)
startup> RUN 100
```

The machine is the Kaleidoscope machine with two changes: the Z80 runs at 2 MHz, and the guest is `GDEMO.HEX`. On a build with **SDL3** the window opens and comes to the front.

Press **STOP** (`Ctrl-E`) to break back to the `altairsim>` prompt. `RUN 100` starts GDEMO again.

GDEMO needs the Z80. The code at `0100` copies the program to `1000` with the Z80 instruction `LDIR` and jumps there.

GDEMO prints nothing and reads no keys on this machine. GDEMO is a CP/M program, and this machine has no CP/M. GDEMO prints its messages with a `CALL` to `0005`, and `GDEMO.HEX` puts a `RET` at `0005`. GDEMO reads its `Ctrl-S` and `Ctrl-C` keys from a Cromemco console at ports `00 / 01`, and this machine has no board there.

The files

File	What it is
<code>kscope.toml</code>	The Kaleidoscope machine.
<code>KSCOPE.HEX</code> , <code>KSCOPE.ASM</code> , <code>KSCOPE.PRN</code>	The program as Intel HEX, its source, and the assembler listing.
<code>dzmbasic.toml</code>	The DZMBASIC machine.
<code>DZMBASIC.DSK</code>	The CP/M disk with <code>DZMBASIC.COM</code> .
<code>DZMBASIC-Manual.pdf</code>	The manual for the <code>DZOP</code> statement and <code>DZF</code> function.
<code>games.toml</code>	The games machine.
<code>GAMES.DSK</code>	The CP/M disk with the Dazzler games.
<code>Cromemco Dazzler Games - Instruction Manual (1978).pdf</code>	The instruction manual for the Cromemco games.
<code>gdemo.toml</code>	The GDEMO machine.
<code>GDEMO.HEX</code>	GDEMO as Intel HEX.
<code>README.md</code> , <code>README.pdf</code>	This file, and the PDF CI builds from it.